

Tiresias

An evidence-backed “5 Whys” for SEC filings: why did the number change, and what should I do about it?

Open Agent Hackathon 2026 · Track: Reasoning Architecture · Palmlore (solo)

“Tiresias named the cause first, but had no evidence and was not believed. This Tiresias brings the evidence.”

1 · THE PROBLEM

3M reported its 2023 revenue twice. The numbers differ by $8 billion.

February 2024

3M's annual report says: **2023 revenue was**

$32.68B

February 2025

The next annual report says: **2023 revenue was**

$24.61B

Same year

$8.07B disappeared. **Why?**

−$8.07B

Companies re-state past years after events like a spin-off. Finding out why means reading several filings, and sometimes another company's.

Source: 3M 10-K filed 2024-02-07 and 10-K filed 2025-02-05 (SEC EDGAR)

Who asks decides where the answer stops

Asker	Levers	Where a restatement question stops
Analyst (primary)	model adjustments, estimates, rating	Re-base the model on continuing operations; the residual is an open question
Auditor	testing scope, follow-up requests, sign-off	Reconciles to the carve-out except the residual: request the elimination schedule
Board	portfolio strategy, capital allocation	Keeps going past the spin-off; often ends at the evidence frontier

Same filings, same evidence, different first actionable cause: the stopping rule is relative to the asker.

2 · THE SOLUTION

Ask “why?” one level at a time, and prove every step

Every link is evidence

Each link in the why-tree carries the SQL sent to the data layer, the rows it returned and the share of the change it explains.

A critic tests each step

Breakdown or cause? Does adding it back reconcile the figures? What is left unexplained (the residual)?

It stops when you can act

At the first cause the asker can act on, with an explicit end state and a recommended action, before or after level five.

End states: actionable · branched · evidence-frontier · external-driver · budget · cyclic (guard)

Every agent action is one JSONL trace line, so a run can be audited after the fact.

2 · ONE RUN, END TO END

3M’s missing $8.07B is Solventum, and the figures reconcile

Loop A · 3M’s own filings

The drop lines up with a disposal: 3M’s 8-K (items 2.01, 9.01) filed 2024-04-04. The amount lives in another registrant, so Loop A stops at the evidence frontier.

Loop B · find and register

EDGAR full-text search finds the filer whose Form 10 names 3M: **Solventum Corp**, a name that shares nothing with 3M’s. You approve; its filings are registered.

Critic · add-back test

```
$24,610M restated
+ $8,197M Solventum
= $32,807M
vs $32,681M original
```

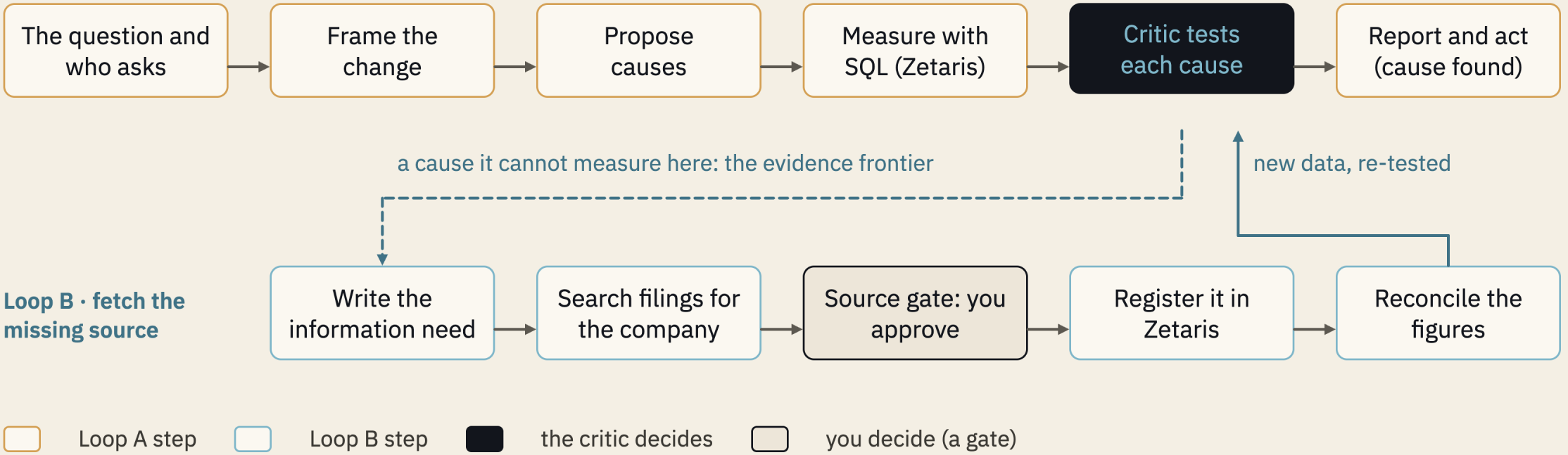
$126M over: 1.6% of the $8,071M change, inside the 5% tolerance. Accepted.

Action (analyst): re-base FY2023 revenue on continuing operations ($24.61B as now reported); the restatement removes Solventum Corp, whose own filings report $8.20B for the period.

output_examples/04 · end state actionable · 5 hops · 7 queries · 1 source registered · same result in sample and live mode

One flow: drill down, and fetch what is missing

Loop A · drill down through the sources already registered



Beside the flow: H-MEM recalls a lesson before causes are proposed, and a memory gate decides whether to store one at the end; Nightshift models may add causes, a second opinion and wording (depth thorough or deep), and never decide.

Bounded: max depth 5, 3 causes per level, 2 new sources per run, 10-minute budget · docs/architecture.md

3 · COLLABORATION IN THE LOOP

Three voices, each with its own job

Tiresias proposes

Frames the change, proposes causes, measures them with SQL, finds missing sources, writes the action.

The Sphinx tests

The critic asks “why?” of every link: cause or breakdown, add-back, residual. It prunes, retries, rejects a lesson that does not fit.

You decide

At the gates: framing, critic escalation, a new source, storing a lesson. Auto, scripted, or live in the UI.

In the web UI’s Truth view, every run counts who did what as it happens (3M case, end of the run, sample mode):

Tiresias proposes	The Sphinx tests	I decide	Zetaris answers
18 steps	6 verdicts	2 gate decisions (0 by me)	0 SQL calls · 13 sample reads

Every gate decision is traced: who decided (you, the policy, or a timeout), the options and the choice · ADR-0006

4 · TECHNICAL STACK

Small services, sponsor tech at the core

Layer	Built with
Agent	Python 3.11 · FastAPI · uv: Loop A and Loop B agents, rule critic, gates, JSONL tracer
Data layer	Zetaris Lightning SQL over REST: SEC EDGAR facts, filings and full-text search as tables
Memory	Meterless H-MEM reference engine (vendored, unmodified) behind a TypeScript REST service
Models	Meterless Nightshift (OpenAI-compatible Runtime) serving NVIDIA Nemotron
Interface	HTTP API (sync, async, server-sent events) and a web UI served by the agent
Ship	Docker Compose (sample mode, no keys) · GitHub Actions CI · Cloud Run demo

Apache-2.0 · contracts/openapi · built with Claude Code (docs/ai-usage.md)

5 · ZETARIS: THE DATA LAYER

Every read is SQL through Zetaris, discovery included

The agent resolves the company through a Zetaris SEC tickers table, reads its facts and filings, searches EDGAR full text, and **registers a newly found company's filings as a Lightning REST table while the run goes on**.

Every claim in a result carries its SQL and the rows returned. No module outside the adapters opens HTTP (a test guards it).

Live run from a fresh Zetaris (trace committed)

Our objects dropped, driver restarted, then one run of the 3M case registered, by itself:

`database · ticker list · 3M facts + views (21 s) · filings · full-text search · Solventum facts (6 s)`

176 trace lines, all live; each call with SQL, outcome, rows, duration, retries. One fallback: the very first read, on an empty Zetaris (logged, B-28).

logs/samples/trace-run-20261024T233153Z-c3227e.jsonl · docs/sponsor-technology.md

5 · ZETARIS: THE SEMANTIC LAYER

A USL over the same SEC sources checks the answer

sec_evidence: three tables, keys and a DQ rule

```
company (cik PK)
filing (accession PK, cik → company)
fact (cik → company, accn → filing)
```

Activated over the 3M and Solventum REST tables the agent registered: 2 companies, 1,262 filings, 539 facts. Rule **period_order**: 0 invalid.

$126M = $126M

One USL query reproduces the agent's residual for 3M's FY2023 add-back.

245 of 539

facts fail the filing key: 3M FY2009–2018 cite filings outside SEC's recent-filings list. The keys also flag 6 filings shared by 3M and Solventum.

29 statements, all run live on 27 Oct and logged. The agent does not read the USL yet; wiring it in is future work.

docs/zetaris-usl.md · zetaris/usl/ · docs/measurements/zetaris-usl-2026-10-27.jsonl

5 · METERLESS AND NVIDIA

A memory engine and the only model provider

H-MEM engine: lessons, not answers

After a run, the generic path that found the cause (no company, figure or date) is stored behind a memory gate. A similar question recalls it and tests that path first; the critic rejects a lesson that does not fit; memory can be purged.

Test: official H-MEM conformance 13/13 · feature: a warm run takes 3 hops instead of 5

Nightshift → NVIDIA Nemotron

Every model call goes to the Nightshift Runtime: extra candidate causes, a second opinion for the critic, the action's wording. On Nightshift 1.2.4 we pin the lanes to Nemotron (Ultra 550B for the critic, Lightning 30B for the fast lane); each trace line records the model that served it and a receipt id.

Committed live run: 4 calls, all Nemotron · **measured 26 Oct:** Ultra 8/8 calls in 1.3–3.2 s; case 04 at depth thorough in 23.5 s

Caveats, stated in the repo: one live run is committed (its hypothesis lane made no call; one reply came back as prose, before JSON mode, and was rejected); the 26 Oct figures are measurements, not committed traces. Models propose; SQL and the rule critic decide.

docs/sponsor-technology.md · services/cognition · adapters/nightshift.py

6 · WHAT WAS UNIQUE ABOUT OUR USE OF THE SPONSOR TECH

A data layer that grows during the run

Loop B finds the missing company from filing evidence, asks the gate, and registers it in Zetaris one statement at a time. Discovery is part of the audited trace.

H-MEM remembers how, never what

A lesson is the generic path that found a cause. The critic judges each recalled lesson in the open; a gate decides what is stored; a purge is ledgered.

Models that never decide, and say who they are

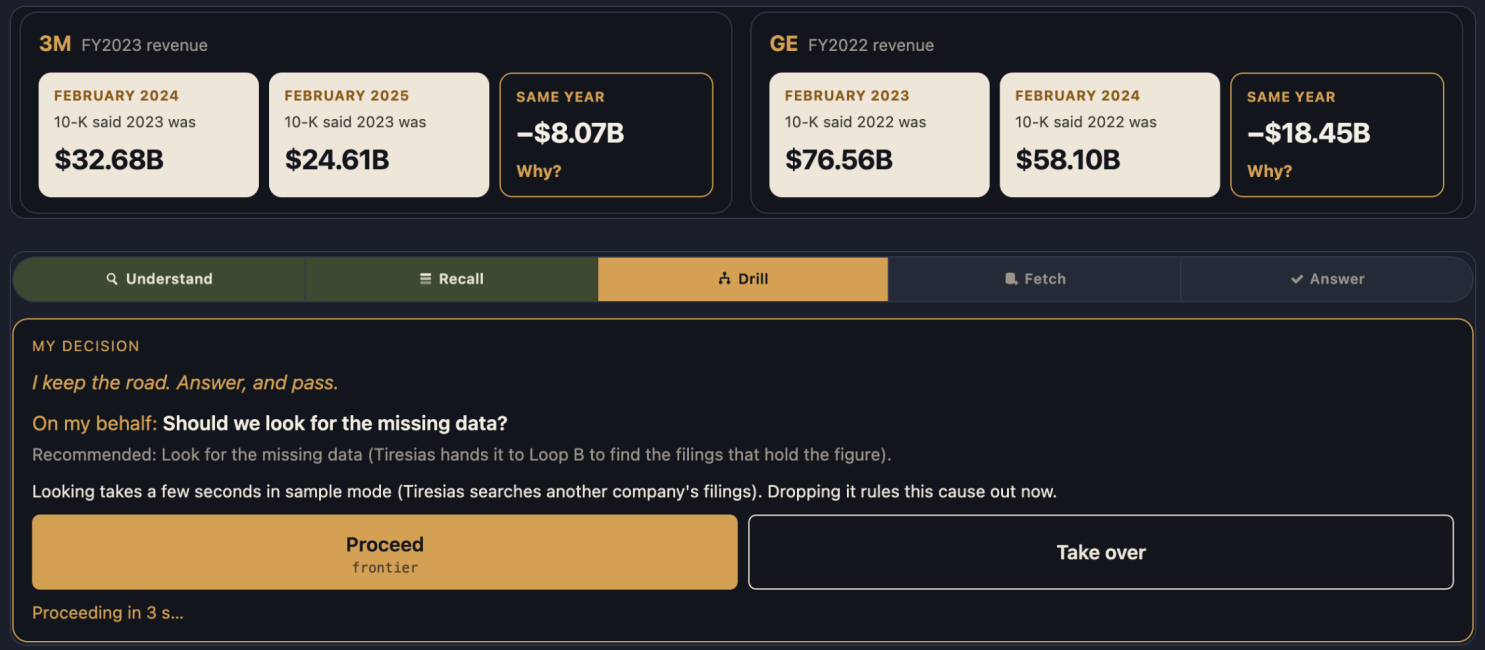
Nemotron through Nightshift proposes; SQL and the rule critic decide. Every call records the model Nightshift actually routed it to.

Stopping relative to the asker

The same why-tree ends at a different first actionable cause for an analyst, an auditor or a board, with human gates in three modes.

docs/sponsor-technology.md · “What was unique”

Steps 1–2 · Ask, then watch it reason

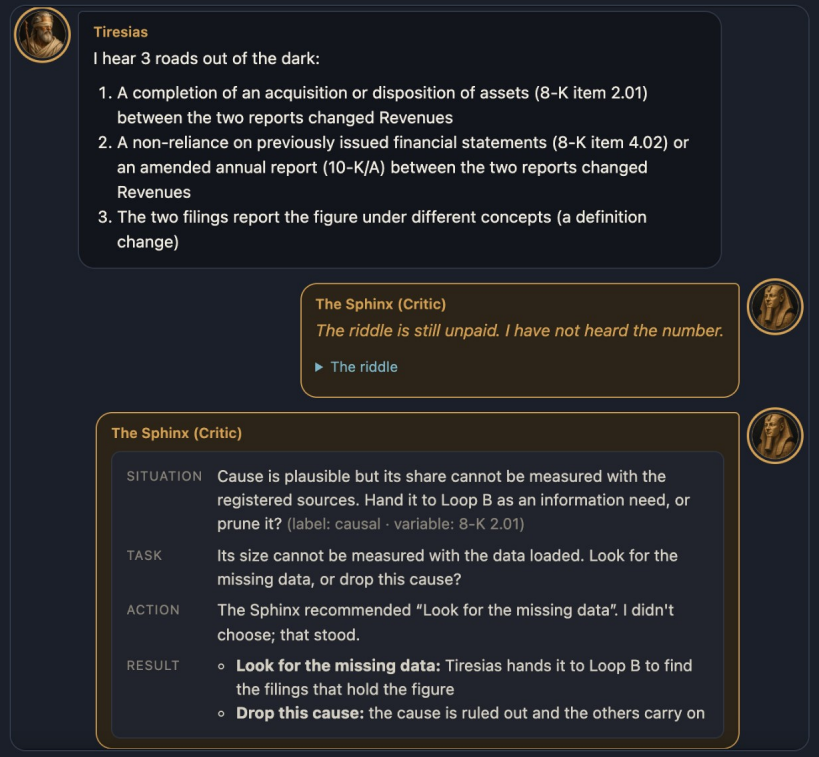


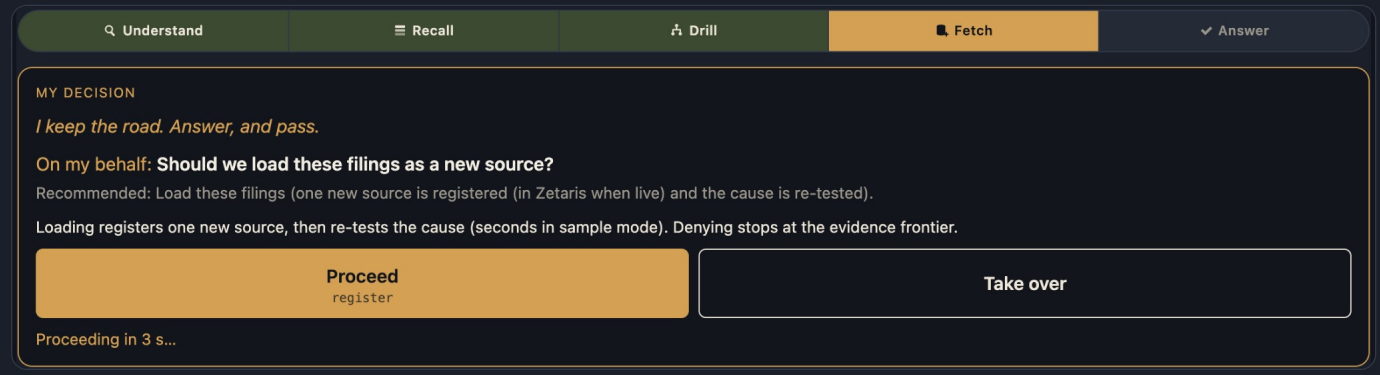
1 · Ask. Each card is a change found in the filings: the same year, reported twice. One click starts the investigation; you can also ask your own.

2 · Watch. The bar fills from Understand to Answer. Tiresias proposes three causes and measures them; then a cause looks right but 3M's own data cannot size it. **Your call**, or the recommendation stands.

Real screenshots of tiresias.palmlore.com, 27 Oct build, sample mode

Steps 3–4 · The Sphinx tests, you decide what it fetches





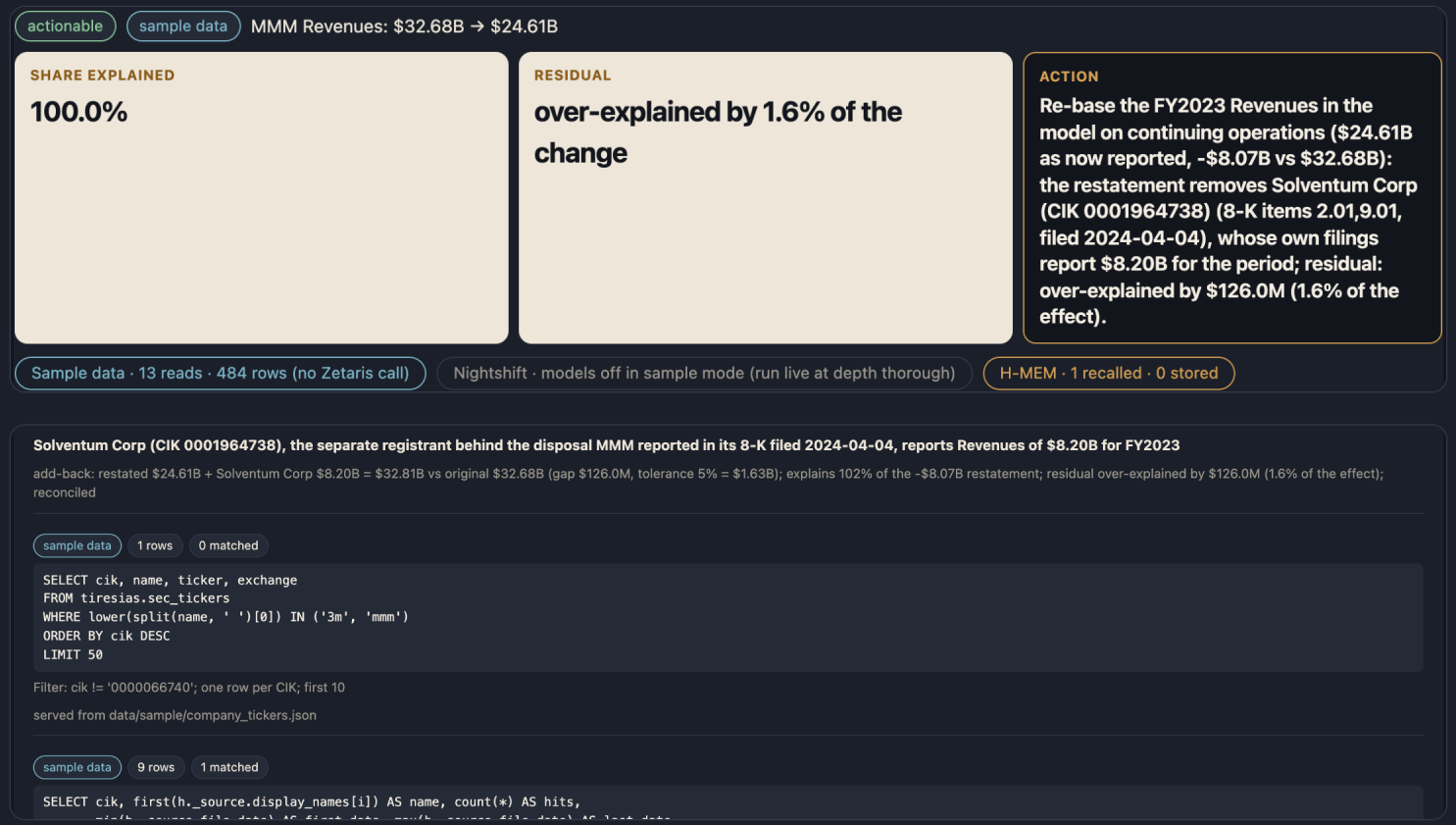
3 · The Sphinx tests. Every answer meets a riddle: is it a cause or a breakdown, and does the number add back? Each decision is laid out as situation, task, action, result.

4 · Approve a new source. Loop B searched the filings and found **Solventum Corp**. Load it? One of at most 2 per run. At the end, a memory gate asks whether to keep the path as a lesson.

Real screenshots, sample mode · every gate shows the recommendation and a deadline; the trace records who decided

7 · USE CASE: AN ANALYST CHECKS 3M'S RESTATED REVENUE

Steps 5–6 · Read the answer, check the evidence

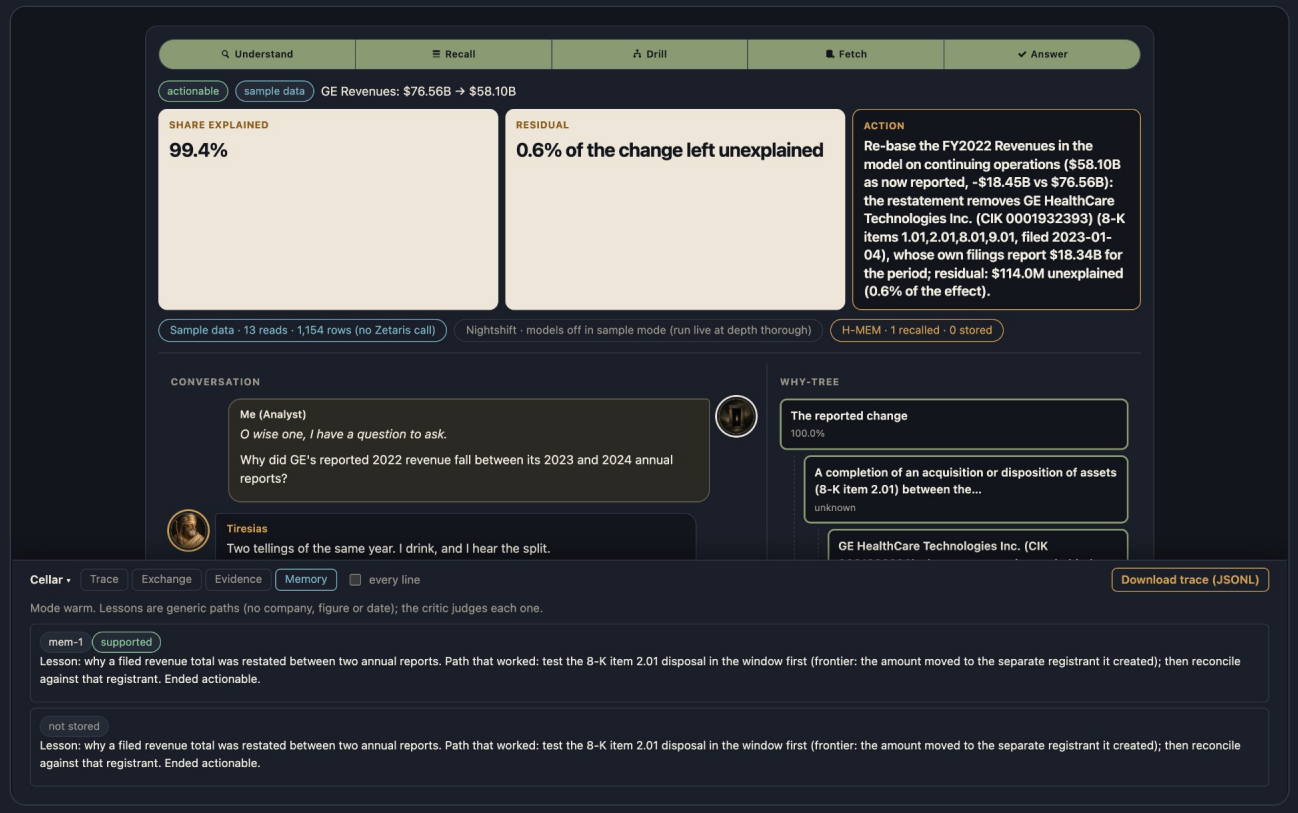


5 · Read. The end state, the share explained, the residual and the action for your role, with chips saying which data, models and memory served the run.

6 · Check. Click a link of the why-tree: the cellar shows the add-back and the SQL and rows it rests on. Truth view shows the same run as raw panels.

Real screenshots, sample mode (registration statements are rendered, not run; live mode runs them in Zetaris)

Step 7 · The next question is faster



Real screenshot, sample mode · the same memory works in live mode

7 · Ask again. Next, GE's restated 2022 revenue. H-MEM recalls the 3M lesson (it holds no company or figure), the Sphinx marks it supported, and Tiresias tests that path first.

5 hops → 3

The same GE question takes 5 hops with empty memory (output_examples/01), 3 with the lesson.

Root cause: GE HealthCare, residual 0.6%. A lesson that does not fit is rejected in the open.

One question in, one action out, in six steps

1 · Ask

Pick a company and a figure, or a suggested question. A what-if or a non-SEC company stops at the scope check, with alternatives.

2 · Recall

Memory recalls the path that worked on a similar past question, and Tiresias tests that path first.

3 · Drill down

Tiresias proposes causes, measures each with SQL, and the Sphinx accepts, prunes or retries it (Loop A).

4 · You decide

Gates wait for you in the UI: approve a new source, settle a model's dissent, keep or skip a lesson.

5 · Fetch what is missing

3M's missing revenue lives in another filer: Tiresias finds Solventum from filings, registers it and re-tests (Loop B).

6 · Act

An end state, the root cause, the residual left unexplained and a recommended action; every link carries its SQL and rows.

7 · COMPONENT BREAKDOWN · THE SPONSORS' VIEW

What each step sends to Zetaris, Meterless and NVIDIA

Step	Zetaris	Meterless	NVIDIA (via Nightshift)
Recall a lesson	—	H-MEM recall from the memory service	—
Propose and measure causes	SQL on its facts and filings (tickers table finds it)	—	Extra candidate causes (fast lane, depth deep)
Test each cause	—	Nightshift Runtime carries every model call	Critic second opinion: Nemotron 3 Ultra 550B
Fetch the missing source	Full-text search, then registers the filer	—	—
Report and remember	—	Memory gate stores the generic path; purge on request	Action wording: Nemotron 3.5 Lightning 30B

Live mode only: every read is Zetaris SQL, and its cache tables are warmed on start. Models run at depth thorough (critic) or deep (all lanes); sample mode makes no model call. Nightshift 1.2.4 honours the model ids we pin.

docs/sponsor-technology.md · docs/agents-reference.md · .env.example (METERLESS_MODEL_FAST / _DEEP)

7 · COMPONENT BREAKDOWN · THE EVIDENCE

How you know each call was made: one trace line per action

A Zetaris call

`sql · rows · duration_ms · retry_count · mode`

Every read and every registration, with the exact SQL sent.

A model call

`lane · requested_model · routed_model · provider · receipt_id`

routed_model comes from Nightshift's receipt, not from our request.

A memory step

`recall · remember · feedback · skip`

Each recall, each stored lesson, and the critic's verdict on a lesson.

Committed: logs/samples/trace-run-20261024T233153Z-c3227e.jsonl, a live run from a fresh Zetaris (176 lines, 4 Nightshift calls); each file in output_examples/ names its trace; scripts/verify_audit.py re-runs the merged PRs' verifiers.

In the UI: each timeline line shows its rows, time, model and receipt; the result chip names the models that served the run. **26 Oct, live:** 3M at depth thorough, 3 of 3 Ultra opinions used, one dissent gate, 23.5 s (PR #39).

Every line carries the run's id (X-Run-Id) · schema: docs/architecture.md#tracing

7 · COMPONENT BREAKDOWN · THE TESTING SETUP

Run it, test it, read its traces

660
agent tests, plus memory service tests

13/13
official H-MEM conformance checks

4 + 2
sample and live outputs, each with its trace

0 keys
to run it: sample mode is the default

CI job	What it checks on every pull request and on main
agent	pytest (loops, critic, gates, Zetaris and Nightshift adapters mocked, API) and ruff
cognition	memory service tests and the official H-MEM conformance suite
repo	no committed secrets; metadata.json valid
quickstart	docker compose up with an empty .env, the web UI served, the four examples through POST /run, then the memory demo (cold, warm, a rejected lesson, purge)

Every change lands as a pull request: an adversarial reviewer (DuelLoop, up to 5 rounds) and green CI before merge. Sample mode runs the real agent logic on a cached SEC slice; nothing is replayed.

.github/workflows/ci.yml · README: Check it yourself · output_examples/ · logs/samples/

8 · FUTURE ENHANCEMENTS

What comes next

Area	Next step
Broader reasoning	More hypothesis rules and cycle detection; the add-back test beyond restatements; branching from any node; narrative text (MD&A, press releases)
Zetaris at scale	One tall Parquet table per company, as Zetaris advised; the agent cross-checks its answer against the USL, and Loop B extends the USL with each source it registers
Live mode in the cloud	Zetaris on a private VM beside Cloud Run; Nightshift on a server once supported
Safer public demo	Load balancer with Cloud Armor: per-visitor rate limits, WAF rules, no direct service URL
Trust signals	A disclosure guardrail for sensitive findings; a run-level flag when live data fell back to the sample



Ask why. Get the evidence.

Try it: tiresias.palmlore.com

Code: github.com/Palmlore/Tiresias

Track: Reasoning Architecture · Palmlore (solo) · Apache-2.0

Built with Zetaris · Meterless H-MEM and Nightshift · NVIDIA Nemotron